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# The OIA Reference Model

*A simple overview of the Open Intelligence Architecture: the ten layers, the seven cross-layer spans, and what they describe about enterprise intelligent systems.*

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## What the OIA is

The Open Intelligence Architecture (OIA) Reference Model describes the structural shape of enterprise intelligent systems. It gives the people who design, deploy, govern, and oversee these systems a shared architectural vocabulary that does not depend on any vendor's product map or any single role's working language.

It is layered, audience-neutral, and additive to the existing reference architectures and management frameworks the enterprise already uses. It is not a product, a methodology for building intelligent systems, or a compliance framework. It is a description of what enterprise intelligent systems are made of, how their layers relate, and what concerns thread through the architecture.

The model is inspired by the OSI reference model that has organised networking for over four decades. The bet is that intelligent systems benefit from the same kind of shared structural vocabulary that networking did — a vocabulary in which vendors, customers, regulators, and boards can all describe the same architecture in the same language.

## The three defining properties of intelligent systems

Three properties distinguish intelligent systems from the application architectures of the past. Each is the reason a specific part of the architecture exists.

**Persistence.** Intelligent systems carry context, memory, and judgment forward across interactions. The persistence is not optional — it is what makes the system useful. It is also what makes it harder to govern.

**Autonomy.** Intelligent systems act with degrees of autonomy that exceed what traditional applications offered. They take multi-step actions, invoke tools, and make decisions inside scopes of authority that humans did not pre-approve case-by-case.

**Consequence.** Outputs and actions whose correctness must be demonstrable — and demonstrable against current threat reality, not against the threat reality at the moment of deployment. Architectural posture is time-bound.

## The ten layers, bottom up

The OIA describes the architecture as ten layers, numbered 0 through 9, consistent with the bottom-up convention of the OSI reference model. Layer 0 is the physical substrate. Layers 1 through 9 are the architectural stack that rests on it.

- **Layer 0 · Physical Compute.** Semiconductors, energy, facilities. The substrate.
- **Layer 1 · Silicon Abstraction.** Hardware diversity behind portable runtimes and trusted execution primitives.
- **Layer 2 · Sovereign Infrastructure.** Compute, storage, and networking with sovereignty made explicit.
- **Layer 3 · Agent Data Substrate.** The persistent, auditable, access-controlled data environment intelligent systems operate on.
- **Layer 4 · Model Training and Adaptation.** Pre-training, fine-tuning, RLHF. The layer at which model weights are created and changed.
- **Layer 5 · Inference and Retrieval.** Where models are run and where retrieval that supports inference is performed.
- **Layer 6 · Context and Knowledge.** Retrieval, grounding, and continuous perception. Where general capability meets specific context.
- **Layer 7 · Orchestration and Workflow.** Coordination of tools, services, and multi-step workflows. The architectural locus of autonomy.
- **Layer 8 · Continuity Fabric.** Cognitive state preserved through change. Recovery, rollback, post-incident integrity. The architectural centrepiece.
- **Layer 9 · Human and Browser Interface.** The surface at which humans and intelligent systems interact.

## The seven cross-layer spans

The cross-layer spans are the surface across which cross-functional governance of intelligent systems is conducted. Where the layers describe where the work happens, the spans describe how the consequences propagate.

- **Security.** Widest scope. Threads through every layer. The named threat surfaces of the agentic era — mimicry, agent-in-the-middle, agent spoofing, model compromise — land at specific layers but require a security posture coherent across the stack.
- **Sovereignty.** Three dimensions: jurisdictional (which legal regime can compel access), operational (whether operating entities are domestic or foreign-controlled), and energy (whether the energy powering the workload is under sovereign supply).
- **Auditability.** The capacity to reconstruct, after the fact, what happened and why. Cannot be retrofitted as a logging layer; must be designed into every layer.
- **Identity.** Three actors require continuous trust governance: humans, agents, and foundation models. The model-as-actor framing is new in v0.5.
- **Energy and Environmental Concerns.** Originating at Layer 0; propagates upward as cost, capacity, and impact at every layer above.
- **Provenance.** The lineage of information, models, decisions, and actions. Without provenance preserved at each layer, auditability fails at every layer above.
- **Verifiability.** New in v0.5. The property that operations can be cryptographically proven rather than contractually asserted. Anchored at Layer 1 by hardware-attested execution primitives.

## The four named threat surfaces of the agentic era

The OIA names four threat surfaces of the agentic era as canonical inside the architecture. The naming gives security audiences a shared vocabulary that ties to the architecture rather than to any specific framework.

- **Mimicry.** Telling humans from agents — and AI from real humans — at every entry point. Originates at Layer 9, propagates through Layer 7, lands in Layer 3 as persisted state.
- **Agent-in-the-middle.** Adversarial intermediaries between agents and tools, including rogue MCP servers — the agentic equivalent of an SSL strip. Concentrates at Layer 7.
- **Agent spoofing.** Agent identities inheriting trust they should not have, across contexts and lifetimes. The inheritance problem at the agent-identity level.
- **Model compromise.** The trusted model that is no longer trustworthy. Weight swap, alignment drift, jailbreak emergence, LLM-specific CVEs. Originates at Layer 5.

Reading the four together surfaces an architectural observation: **three of the four originate or propagate through Layer 7, and the fourth originates at Layer 5.** The security architecture of the agentic era concentrates its weight inside the operational stack — not at the perimeter where prior security architectures expected attacks to land.

## How the OIA fits with existing frameworks

The OIA is additive, not competitive. It addresses a domain that existing reference architectures and management frameworks were not designed to organise, while drawing on their intellectual conventions and complementing their coverage. The companion architecture map shows each OIA layer with the existing framework references that apply to it. A summary:

- **NIST Cybersecurity Framework (CSF).** A functional framework describing cybersecurity activities. The CSF's five functions — Identify, Protect, Detect, Respond, Recover — apply at every OIA layer.
- **NIST AI Risk Management Framework.** Organises how the enterprise manages AI risk. The OIA organises the architecture of the systems the AI RMF asks the enterprise to manage.
- **MITRE ATT&CK and ATLAS.** Threat taxonomies. The OIA provides structural vocabulary within which threats locate; ATLAS provides threat vocabulary that informs what must be defended at each layer.
- **MITRE D3FEND.** Defender-side taxonomy of techniques. Locates at the OIA layers where its techniques apply.
- **OWASP Top 10 for LLM Applications.** A risk enumeration. The OWASP LLM risks manifest at specific OIA layers (notably Layers 5, 6, and 7) and inform the key concerns of those layers.
- **Confidential Computing Consortium.** Provides the hardware-attested execution primitives that anchor the Verifiability span at Layer 1 and operationalise at Layers 2, 4, 5, and 8.
- **ISO/IEC 42001.** An AI management system standard. Operates at management-system level; the OIA operates at architectural level. Organisations certified against 42001 benefit from the structural vocabulary the OIA provides.

*A practical composition: OSI for networking, ISO/IEC 42001 for management systems, NIST CSF for cybersecurity programme structure, NIST AI RMF for AI risk governance, MITRE ATLAS and OWASP LLM Top 10 for threat awareness, and the OIA for the architecture of the intelligent systems themselves.*

## Where to read more

This overview is the brief. The full body of work at v0.5 includes:

- **The OIA Reference Paper.** The architectural specification in full — roughly forty pages. The authoritative reference.
- **The OIA Lens System.** Three audience-specific Lens Maps. The CISO Lens is published with this overview; the CIO/CTO Lens and the Board and Risk Lens are available separately.
- **The OIA Visual Reference.** The three architectural diagrams — the Stack, the Lens Overlay, and the Four Named Threat Surfaces — with explanatory captions.
- **The OIA Canonical Knowledge Base.** A single-source-of-truth Markdown document consolidating the body of work for retrieval-based applications.